RGPV B.Tech FIRST YEAR - SEMESTER 1 (AI/ML / SET-B)

2026 EXAM PREDICTION REPORT

Most Probable Questions + % Chance (Mid Sem 1 + End Sem)

Data: 6 actual RGPV university papers analyzed question-by-question - June 2023, Dec 2023, Dec 2024, June 2025, Dec 2025, June 2026 (latest). % = historical repetition rate (kitne papers mein aaya). Prediction guarantee nahi hai, lekin RGPV ka question-bank pattern highly stable hai - 100% wale topics chhodna manaa hai!

100% = har paper mein aaya (6/6 ya 5/5) | 80% = lagbhag har baar (4/5-5/6) | 60% = due/rotation - wapas aa sakta hai

Exam pattern: 3 hrs, 70 marks - 8 mein se KOI BHI 5 questions attempt (har Q = 14 marks, a+b). [MS1] = Mid-Sem-1 zone (early units - mid sem ke liye yahi karo).

BT-102 - MATHEMATICS-I (M1) | 5 papers analyzed

100% CHANCE - INHE CHHODNA MAT (har paper mein aaya)

MOST PROBABLE QUESTION	% CHANCE	KAB-KAB AAYA (papers)
$u = \log(x^3+y^3+z^3-3xyz)$, prove: $(\partial/\partial x + \partial/\partial y + \partial/\partial z)^2 u = -9/(x+y+z)^2$	**100%**	Dec 2025 + June 2026 - back-to-back SAME question! Is baar bhi sabse zyada chance
Maxima/minima of x^3+y^3-3axy (variants: $x^3+y^3-63(x+y)+12xy$, rectangular solid max volume in sphere = cube)	**100%**	Saare 5 papers. Cube-in-sphere 3 BAAR repeat ho chuka
Taylor/Maclaurin series expansion (sin x at π/3 ya π/2; $e^x \cos y$; e^{xy} at (1,1); x^4+x-2 at a=1)	**100%**	Saare 5 papers
Beta-Gamma functions: prove $B(m,n) = \Gamma(m)\Gamma(n)/\Gamma(m+n)$ + numerical evaluate ($\int x^2(1-x)^3\,dx$ type)	**100%**	Saare 5 papers - har paper mein 2 questions
Fourier series ($f(x)=x^3$, x^2, $x+x^2$, x sin x; half-range sine of x(π-x))	**100%**	Saare 5 papers - often 2 questions

80% CHANCE - YE BHI PAKKE PAKDE JAYEIN

MOST PROBABLE QUESTION	% CHANCE	KAB-KAB AAYA (papers)
Rolle's / Lagrange theorem verify (latest: $f(x)=e^x \sin x$ in [0, 2π])	**80%**	4/5 papers
Eigenvalues & eigenvectors - sirf 2 standard matrices aate hain: [[8,-6,2],[-6,7,-4],[2,-4,3]] (Dec24+Dec25) & [[2,-2,2],[1,1,1],[1,3,-1]] (Dec23+Jun25)	**80%**	4/5 papers - dono matrices ratt lo
Change of order of integration	**80%**	4/5 papers
Convergence test of series: $1 + x/2 + x^2/5 + x^3/10 + \ldots$	**80%**	Dec 2023 + Jun 2026 SAME series
Rank of matrix by normal form (echelon)	**80%**	4/5 papers
Consistency & solve system / λ-μ wala question (no/unique/infinite solutions)	**80%**	4/5 papers
Area of cardioid / parabolas, volume of revolution	**80%**	4/5 papers

60% - DUE / ROTATION PICKS

MOST PROBABLE QUESTION	% CHANCE	KAB-KAB AAYA (papers)
Diagonalize matrix / Cayley-Hamilton theorem + find A^{-1}	60%	Jun 2026 mein Cayley-Hamilton aaya - ab diagonalization due
Subspace prove karna / vectors linearly dependent ya independent	60%	2 papers skip ho chuke - wapas aana chahiye

Mid-Sem-1 pack (Maths): Rolle's theorem verify, Taylor series, $u=\log(x^3+y^3+z^3-3xyz)$ partial derivative, maxima-minima x^3+y^3-3axy + box/cube problem.

BT-201 - ENGINEERING PHYSICS | 6 papers analyzed

100% CHANCE - GUARANTEED LEVEL (6/6 papers)

MOST PROBABLE QUESTION	% CHANCE	KAB-KAB AAYA (papers)
Particle in 1D box - energy eigenvalues + wavefunction + Time-Independent Schrödinger Equation derivation	100%	SAARE 6 papers! June 2026 mein dono ek saath aaye
Optical fiber - NA & acceptance angle derivation + numerical (RI 1.55/1.50, 1.75/1.70, 1.45/1.41 type)	100%	SAARE 6 papers - numerical pakka aata hai

67-83% CHANCE

MOST PROBABLE QUESTION	% CHANCE	KAB-KAB AAYA (papers)
Fermi level in intrinsic & extrinsic semiconductors (diagram ke saath)	83%	5/6 papers
Heisenberg uncertainty principle - state + elementary proof	67%	4/6 papers
Newton's rings - theory + numerical (6000 Å, 10th dark ring, R = 100 cm type)	67%	4/6 papers
Rayleigh criterion of resolution + resolving power of grating	67%	4/6 papers
Hall effect - Hall voltage / Hall coefficient derivation	67%	4/6 papers
Einstein A & B coefficients + population inversion	67%	4/6 papers
Continuity equation from charge conservation (same wording Dec24 + Jun25)	67%	4/6 papers

50% - STRONG SUPPORT

MOST PROBABLE QUESTION	% CHANCE	KAB-KAB AAYA (papers)
Michelson interferometer (fringe shift derivation)	50%	3/6 - last Dec 2025
He-Ne laser construction & working	50%	3/6
CO_2 laser / Ruby laser construction & working	50%	Jun26 + Jun25 mein CO_2
Bloch theorem / Kronig-Penney model (short note)	50%	3/6
Stokes & Gauss divergence theorem	50%	3/6
YDSE - Young's double slit + fringe width	50%	3/6
V-number / wave packet / Vp-Vg relation	50%	rotation

Due (wapas aa sakte hain): Mach-Zehnder interferometer, Fraunhofer single-slit diffraction, Zener/P-N diode V-I characteristics, solar cell, density of states.

Mid-Sem-1 pack (Physics): Particle in box FULL derivation, Schrödinger TISE, uncertainty principle + numerical, Newton's rings derivation + numerical, Rayleigh criterion, YDSE fringe width.

BT-203 - BASIC MECHANICAL ENGINEERING (BME) | 5 papers analyzed

100% CHANCE - har paper mein Q1 se yahi shuru hota hai

MOST PROBABLE QUESTION	% CHANCE	KAB-KAB AAYA (papers)
Engineering materials - alloys, alloy steels, cast iron classification + applications	**100%**	5/5 papers [MS1]
Mechanical properties (toughness, brittleness, ductility, strength) + tensile test + stress-strain curve of ductile material	**100%**	5/5 papers [MS1]
Measuring instruments - vernier caliper, micrometer, sine bar, optical pyrometer (construction + least count)	**100%**	5/5 papers [MS1]
Lathe main parts (sketch) / drilling machine working	**100%**	5/5 papers [MS1]
Thermodynamics - open/closed/isolated systems + Zeroth law / First law	**100%**	5/5 papers
IC engines - 4-stroke petrol/diesel (sketch), 2-stroke compare, steam engine	**100%**	5/5 papers

80% CHANCE

MOST PROBABLE QUESTION	% CHANCE	KAB-KAB AAYA (papers)
Bernoulli's equation derivation + pipe numerical (30/80 lit/sec wala due hai)	**80%**	4/5 papers
Newton's law of viscosity + flow types (laminar/turbulent/Newtonian)	**80%**	4/5 papers
Boilers - classification / mountings vs accessories / Babcock-Wilcox	**80%**	4/5 papers
Compressors - reciprocating working / centrifugal short note	**80%**	4/5 papers

40-60% - ROTATION (Iron-Carbon GARAM hai!)

MOST PROBABLE QUESTION	% CHANCE	KAB-KAB AAYA (papers)
Iron-Carbon diagram - phases, critical points	**40% → HOT**	Dec 2025 + June 2026 back-to-back! Is baar bhi possible
Hydraulic turbines - impulse vs reaction, Pelton wheel	**60%**	3/5
Reciprocating pump + priming / centrifugal pump	**60%**	3/5
2nd law - Kelvin-Planck & Clausius statements	**40%**	2/5 - due
Diesel/Dual cycle efficiency derivation	**40%**	2/5
Draught - natural & artificial (short note)	**60%**	3/5

Mid-Sem-1 pack (BME): Materials classification + alloys, 4 mechanical properties definitions, tensile test/stress-strain, hardness test (Brinell - due hai!), vernier/micrometer/sine bar, lathe + drilling.

BT-204 - BASIC CIVIL ENGINEERING & MECHANICS | 5 papers analyzed

100% CHANCE

MOST PROBABLE QUESTION	% CHANCE	KAB-KAB AAYA (papers)
Concrete - workability, compaction, curing, lab tests, nominal/design mix	100%	5/5 papers [MS1]
Remote sensing (components + applications) & GPS positioning	100%	5/5 papers [MS1]
Building components - stairs, doors/windows, footings (spread/RCC), plastering/pointing	100%	5/5 papers [MS1]

80% CHANCE

MOST PROBABLE QUESTION	% CHANCE	KAB-KAB AAYA (papers)
Levelling - RL numerical (Rise-Fall method Dec24, HI method Jun26) + reciprocal levelling	80%	Numerical pakka aata hai
Bricks/stones - manufacturing, properties, tests + masonry & mortar	80%	Last 3 papers strong trend
Theodolite / electronic surveying / traversing methods	80%	4/5
Contours - properties + methods of contouring	80%	4/5
Truss - method of joints/sections numerical	80%	4/5
SFD & BMD - SSB with UDL + point load (classic)	80%	4/5
Centroid vs CG / MOI / parallel axis theorem / radius of gyration	80%	4/5
Force system, resultant, FBD, D'Alembert principle	80%	4/5

40-60% - DUE PICKS

MOST PROBABLE QUESTION	% CHANCE	KAB-KAB AAYA (papers)
Friction - laws + inclined plane numerical	60%	due hai
Area computation numerical - Simpson's/trapezoidal/mid-ordinate rule	60%	due hai
Cement - composition, field/lab tests	40%	2/5
Stress - elongation of rod numerical	40%	2/5
Chains & tapes - distance measurement	40%	2/5

Mid-Sem-1 pack (BCEM): Concrete (workability, curing, compaction), bricks/stones, building components (stairs/doors/foundations), remote sensing + GPS, contours, survey basics.

BT-205 - BASIC COMPUTER ENGINEERING (BCE) | 5 papers analyzed

100% CHANCE

MOST PROBABLE QUESTION	% CHANCE	KAB-KAB AAYA (papers)
OSI model - 7 layers + functions, TCP/IP model, dono ka comparison	100%	5/5 papers
Cloud computing - IaaS/PaaS/SaaS service models + cloud infrastructure	100%	5/5 papers

MOST PROBABLE QUESTION	% CHANCE	KAB-KAB AAYA (papers)
DBMS - architecture, DBA roles, primary key, data dictionary, vs file system	**100%**	5/5 papers
C++ - constructors (types) + friend function (example ke saath)	**100%**	5/5 papers
OOP characteristics / POP vs OOP comparison	**100%**	5/5 papers [MS1]
Algorithm + flowchart - LARGEST OF 3 NUMBERS (3 BAAR repeat!)	**100%**	Dec24 + Jun25 + Jun26 [MS1]

80% CHANCE

MOST PROBABLE QUESTION	% CHANCE	KAB-KAB AAYA (papers)
Memory - primary/secondary, memory hierarchy, storage + I/O	**80%**	4/5 [MS1]
Operating system - functions, types, kernel, file/process/memory management	**80%**	4/5
Computer classification (super/mainframe/embedded) + applications in fields	**80%**	4/5 [MS1]
C++ programs (swap 2 numbers), operators, scope resolution, data types	**80%**	4/5
Security - virus/worm/trojan/malware + firewall + cyber law/ethics	**80%**	Jun25 + Jun26 garam

40-60% - RISING TOPICS

MOST PROBABLE QUESTION	% CHANCE	KAB-KAB AAYA (papers)
Inheritance (types) + polymorphism (overloading, virtual functions)	**60% → HOT**	June 2026 mein dono aaye - trend strong
MS-Word/Excel/PowerPoint features	**40% → rising**	Jun 2026 mein wapas aaya
CPU organization - registers, bus architecture	**40%**	2/5
System vs application software	**60%**	3/5 [MS1]

Mid-Sem-1 pack (BCE): Computer fundamentals + classification, software types, memory types, OS basics, algorithm/flowchart (largest-of-3 likhna aana hi chahiye), OOP characteristics, C++ data types.

GOLDEN LIST - WORD-TO-WORD REPEAT HO CHUKE QUESTIONS (sabse zyada chance)

#	EXACT QUESTION	KAB-KAB AAYA
1	$u = \log(x^3+y^3+z^3-3xyz)$, prove $(\partial x+\partial y+\partial z)^2 u = -9/(x+y+z)^2$	Dec 2025 + June 2026
2	Eigenvalues of [[8,-6,2],[-6,7,-4],[2,-4,3]]	Dec 2024 + Dec 2025
3	Eigenvalues of [[2,-2,2],[1,1,1],[1,3,-1]]	Dec 2023 + June 2025
4	Series test: $1 + x/2 + x^2/5 + x^3/10 + \ldots$	Dec 2023 + June 2026
5	Half-range sine series of $x(\pi-x)$	June 2025 + June 2026
6	Fourier: $f(x) = x \sin x$ expansion	Dec 2023 + June 2026
7	Rectangular solid of max volume in sphere = cube	Dec 2023 + Dec 2024 + June 2025 (3 baar!)
8	Maxima/minima of x^3+y^3-3axy	Dec 2024 + Dec 2025

#	EXACT QUESTION	KAB-KAB AAYA
9	Continuity equation from charge conservation (same wording)	Dec 2024 + June 2025
10	Flowchart/algorithm - largest of 3 numbers (BCE)	Dec 2024 + June 2025 + June 2026 (3 baar!)
11	Fiber NA numerical (core/cladding RI diye hue)	Jun 2023 + Dec 2023 + Dec 2024 + Dec 2025
12	Particle in 1D box - eigenvalues + wavefunction	SAARE 6 papers!
13	OSI 7-layers + TCP/IP (BCE)	5/5 papers - in dono mein se ek har baar

FINAL STRATEGY - 80/20 RULE (time kam hai to sirf ye karo)

1. MATHS: u=log(...) partial derivative, x^3+y^3-3axy maxima, Taylor series, Beta-Gamma (relation + 1 numerical), Fourier (x^3/x^2/half-range), eigenvalues (dono standard matrices), Rolle's verify.

2. PHYSICS: Particle in box FULL derivation, fiber NA (derivation + numerical), Newton's rings, uncertainty principle, Fermi level diagram, Hall effect, Einstein A&B, He-Ne laser.

3. BME: Materials + alloys (Q1 pakka), stress-strain curve, vernier/sine bar/micrometer, lathe sketch, thermo systems + Zeroth law, 4-stroke engine, Bernoulli.

4. BCEM: Concrete, remote sensing/GPS, levelling numerical (Rise-Fall + HI dono), truss joints method, SFD/BMD (SSB+UDL), centroid vs CG.

5. BCE: OSI vs TCP/IP, cloud IaaS/PaaS/SaaS, DBMS+DBA, constructor/friend function, OOP characteristics, largest-of-3 flowchart, memory types, virus/malware.

EXAM HALL TIP: 8 mein se sirf 5 attempt karne hote hain. Upar ke topics se tumhe har unit mein 1-1 strong option mil jayega - 100%/80% wale questions mein se 2-3 options har paper mein pakka milte hain. Numericals (fiber NA, Newton's rings, RL, truss, eigenvalues) PRACTICE karo, sirf ratte se kaam nahi chalega. All the best!

Source: RGPV actual question papers June 2023, Dec 2023, Dec 2024, June 2025, Dec 2025, June 2026 (verified PDFs - rgpvonline.com). Ye frequency-analysis based prediction hai, official RGPV document nahi. Compiled: October 2026.